APPLIED SCIENCES AND ENGINEERING

Machine learning–enhanced molecular network reveals global exposure to hundreds of unknown PFAS

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Unknown forever chemicals like per- and polyfluoroalkyl substances (PFASs) are difficult to identify. Current platforms designed for metabolites and natural products cannot capture the diverse structural characteristics of PFAS. Here, we report an automatic PFAS identification platform (APP-ID) that screens for PFAS in environmental samples using an enhanced molecular network and identifies unknown PFAS structures using machine learning. Our networking algorithm, which enhances characteristic fragment matches, has lower false-positive rate (0.7%) than current algorithms (2.4 to 46%). Our support vector machine model identified unknown PFAS in test set with 58.3% accuracy, surpassing current software. Further, APP-ID detected 733 PFASs in real fluorochemical wastewater, 39 of which are previously unreported in environmental media. Retrospective screening of 126 PFASs against public data repository from 20 countries show PFAS substitutes are prevalent worldwide.

INTRODUCTION

Per- and polyfluoroalkyl substances (PFASs) are man-made chemicals used to create various industrial and consumer products that resist heat, oil, and water (*1*). These chemicals with strong carbon-fluorine bonds are known as "forever chemicals" because they persist in the environment (*2*, *3*). Metabolomics and exposome research have shown that PFAS exposure can cause liver and kidney damage, developmental problems in fetuses and infants, and increase the risk of certain types of cancers (*4*–*6*), leading several countries to ban or restrict their use (*2*). Banning legacy PFAS, however, has led to the increased use of PFAS substitutes that have not been adequately tested for safety (*7*) or publicly disclosed due to trade secret protection (*8*). Environmental transformations (*9*) of these chemicals also mean there are many more unknown PFAS in the environment that are not on the list maintained by the US Environmental Protection Agency (EPA) (*10*) and need to be identified urgently given their similar persistence and toxicity to legacy PFAS (*11*, *12*).

Nontarget screening based on high-resolution mass spectrometry (HRMS) has been used to identify unknown PFAS among thousands of features in complex environment samples. Similar to analysis of lipids or metabolites, this method assumes that the compounds occur as homologs or contain specific PFAS-related fragments (*13*–*16*). While this knowledge-driven approach has successfully identified unknown classes of PFAS in environmental and human blood samples (*17*, *18*), it is unable to capture the increasingly diverse structural characteristics of unknown PFAS, leaving many of them unidentified. Moreover, structural annotation of these unknown PFAS currently relies heavily on expert experience, and manual identification is time-consuming and tedious. Having an automated data-driven PFAS identification platform is thus crucial for expediting the process.

Molecular network is a promising data-driven nontarget screening method for discovering unknown compounds (*19*–*21*). Implemented on several metabolite and natural product identification platforms such as global natural products social molecular networking (GNPS) and metabolite identification and dysregulated network analysis (*22*), this method creates a peak node network that is linked together by mass spectra similarity. Because the method assumes that pairs of highly similar mass spectra share similar structures, the number of identified compounds and the likelihood of discovering unknown compounds increases (*21*). The GNPS platform has demonstrated efficacy in identifying a wide range of exposome chemicals, such as antimicrobial and pharmaceutical transformation products (*23*, *24*). Nonetheless, constructing a molecular network relies on spectra similarity algorithms. Given the rapid development of spectra similarity algorithms, such as entropy (*25*) and Bonanza (*26*), it remains unclear which is best suited for PFAS identification considering identification throughput and accuracy. Moreover, the grouped spectra are identified using CFM-ID (*27*), MetFrag (*28*), or SIRIUS (*29*) that source their training sets from databases like GNPS, NIST 17, and MassBank. The effectiveness of these platforms in identifying PFAS remains uncertain due to infrequent presence of PFAS in these datasets. Further, despite the registration of 7 million PFAS compounds in PubChem (*30*), many PFAS compounds produced remain undisclosed due to trade secret protection (*8*), and numerous transformation products of PFAS are absent from structure databases (*9*). In addition, only a tiny fraction has associated MS spectra available for the training of current models. Therefore, any platform for accurately identifying unknown PFAS in environmental samples will require more PFAS MS/MS data and alternative identification strategies.

Here, we report a PFAS identification platform that can discover and accurately identify unknown PFASs in environmental samples. Our automatic per- and polyfluoroalkyl identification (APP-ID) platform contains a PFAS-only molecular network (called PFAS_link) and a PFAS identification module (PFAS_ID) that is trained using a PFAS database we built in-house. To reduce false-positive linkages in the PFAS_link network, we developed a Flink algorithm that removes common nonspecific fragment or neutral loss matches. Unlike traditional identification platforms that only include exact candidates, our platform uses exact and transformation searches to identify both known PFAS and unknown ones that are not in the structure database. Analyzing real wastewater samples obtained from a fluorochemical industrial park in Changshu, Jiangsu Province, China with APP-ID identified 31 classes of PFAS, 17 of which are previously unknown classes. Our results show that APP-ID is effective and

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suggest that unknown PFAS contamination is as widespread as legacy PFAS.

RESULTS

Overview of APP-ID

Our APP-ID workflow starts with MS data whose peak table contains the peaks' mass/charge ratio (*m*/*z*), retention time, intensity, and MS1 and MS2 spectra (Fig. 1). Within the workflow, PFAS_link module filters non-PFAS features and provides only PFAS features for PFAS_ID to identify the structures. Specifically, the PFAS structures were iteratively annotated according to the loop-based workflow, and one round stands for the annotation process from seeds (known PFAS) to neighbor (unknown PFAS). First, sample MS data are compared with a PFAS spectra library to obtain the first batch of seeds, and we named this process as round 0. In the next each round, the neighbors were screened using Flink and then annotated the structure using PFAS-ID. After each round, the annotated neighbors were selected as the seeds for the next round. If there is no newly identified PFAS for neighbors, this workflow is finished. Round number represents the number of iterations from PFAS spectra library in APP-ID. To annotate these tentative PFAS features, the PFAS_ID module retrieves structure candidates from exact and transformation searches, uses a machine learning model to predict fingerprints of PFAS features from MS2 spectra, and ranks the candidates from both searches according to fingerprint similarity. Structure candidates are retrieved from PubChem and an in-house PFAS database (6484 compounds). This retrieval is performed under a relative mass error of 5 parts per million (ppm) and an isotope distribution fit threshold of 0.8 using the IsoSpecPy Python package (*31*, *32*), as these thresholds were validated by PFAS standards (fig. S1).

Because current identification platforms such as CFM-ID (*27*) and MetFrag (*28*) cannot identify unknown PFAS that are not in the structure databases, we developed a transformation search algorithm to aid manual structure annotation. Assuming that unknown PFAS come from the transformation of known PFAS, structure candidates are obtained from the exact mass of known PFAS and mass shifts between known and unknown PFAS features. Our transformation search algorithm, which returns candidates with reasonable mass shifts from known PFAS in our in-house PFAS database, allows us to deduce a structure for the unknown PFAS.

To account for structural changes of known PFAS after transformation, ranking scores of candidates from the transformation search are degraded based on the elemental composition of mass shift. Newly identified PFAS obtained from the PFAS_ID module are added into seed PFAS to iteratively screen for more PFAS. Adducts are annotated to remove redundant seeds.

Constructing the PFAS_link module

We built a PFAS-only molecular network using a self-developed Flink algorithm that improves connections between PFAS and excludes non-PFAS (compounds without fluorine). To construct a PFAS-only molecular network, our Flink algorithm introduces three revisions based on the conventional dot product (DP) algorithm (*33*). First, our Flink algorithm uses fragment peak matching ratio to weaken the influence of peak intensity because the fragment intensity varies hugely even for PFAS with similar structures (fig. S2). Second, we incorporate neutral loss matching to find PFAS with diverse structures. Last, we have excluded common fragments or neutral losses to diminish false-positive results. A detailed description of the Flink algorithm, including its equation, was provided in Material and Methods.

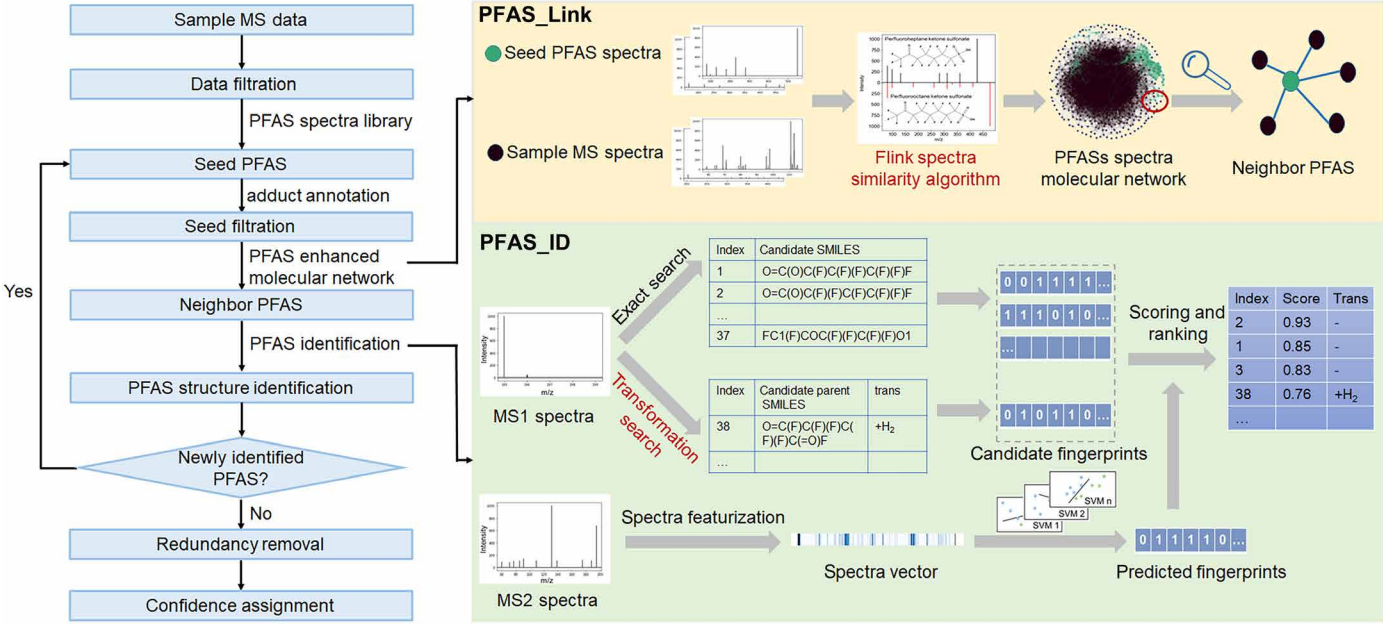


Fig. 1.The workflow diagram for the APP-ID method. The APP-ID method workflow mainly includes (i) import sample MS data, (ii) seed PFAS annotation, (iii) PFAS_link module that searches for neighbor PFAS, (iv) PFAS_ID module that identifies the structure of PFAS, (v) iterative identification, and (vi) redundancy removal and confidence assignment.

We disconnected non-PFAS in the network by excluding common neutral loss or common fragment match that could exist in the spectra of both PFAS and non-PFAS. To illustrate the impact of excluding common neutral losses and common fragments, we compared Flink under three scenarios: scenario 1, without any treatment; scenario 2, with common neutral losses excluded; and scenario 3, with both common neutral losses and common fragments excluded. For example, indoxyl sulfate (a non-PFAS) has three fragment peaks that closely match PFOS (a PFAS); one fragment is directly matched by m/z and two fragments are matched by neutral loss (Fig. 2A). Excluding common neutral loss matches (parent ion 0, CO_2 and SO_3) in scenario 2 leaves only one matched fragment. Additional exclusion of fragments with $m/z < 100$ (except for PFAS characteristic ions, SO_3F^- and SO_2F^-) in scenario 3 brings the Flink score to 0.0, which disconnects indoxyl sulfate and PFOS at any similarity threshold.

We also evaluate the performance of our Flink algorithm under three scenarios with 60 PFAS and 700 non-PFAS spectra. In our evaluation, we computed the spectral similarity across all 760 spectra, linking compound pairs with similarity scores surpassing spectra similarity threshold. The positive rate denotes the percentage of PFAS successfully linked with other PFAS in the molecular network, while the false-positive rate indicates the percentage of non-PFAS linked with PFAS in the molecular network. After common neutral loss and fragment exclusion, scenario 3 drops sharply in the false-positive rate across the threshold range of 0 to 1, while maintaining a consistent positive rate (Fig. 2B). Specifically, under a similarity threshold of 0.5, the false-positive rate decreases from 35 to 2.5%

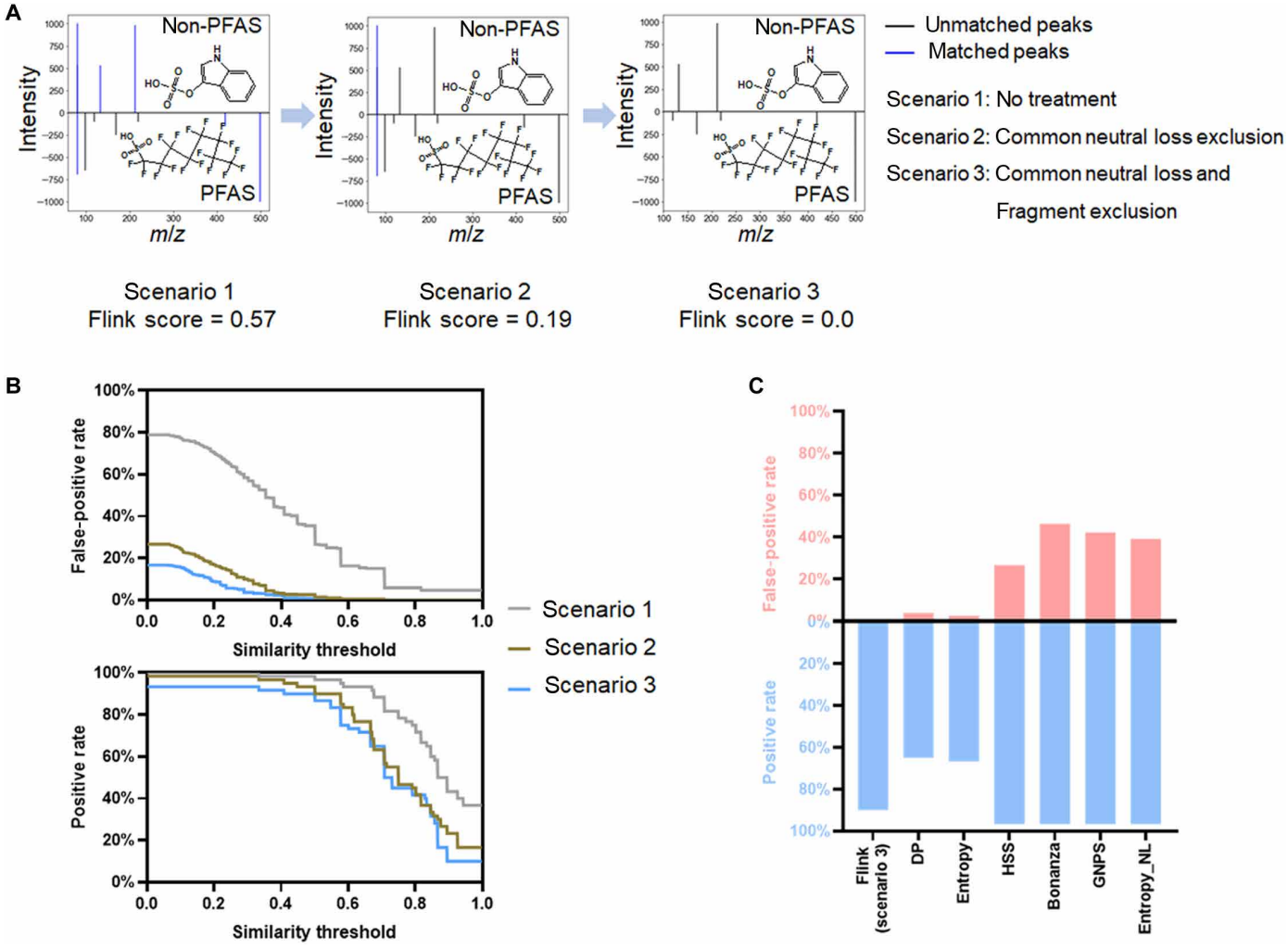


Fig. 2. Flink algorithm outperform traditional algorithms in PFAS molecular network. (**A**) An example of Flink algorithm in distinguishing PFAS with non-PFAS under three scenarios. Scenario 1 represents the baseline without any treatment, while scenario 2 involves the exclusion of common neutral losses, including 0, CO_2, and SO_3. In scenario 3, common neutral losses and common fragments below 100, except for PFAS characteristic ions SO_3F^- and SO_2F^-, are excluded. (**B**) False-positive rate and positive rate of Flink algorithm under similarity threshold 0 to 1 across three scenarios using 60 PFAS spectra and 700 non-PFAS spectra. (**C**) False-positive rate and positive rate of seven spectra similarity algorithms under a similarity threshold of 0.5 using 60 PFAS spectra and 700 non-PFAS spectra. Compound pairs that have spectra similarity higher than similarity threshold were linked and positive rate stands for percentage of PFAS that could be linked with other PFAS in molecular network; false-positive rate for percentage of non-PFAS that could be linked with PFAS in molecular network.

after common neutral loss exclusion in scenario 2 and further decreases from 2.5 to 0.7% after common fragment exclusion in scenario 3. Meanwhile, the positive rate remains at 90% in scenario 3. Therefore, we used Flink in scenario 3 for constructing the PFAS molecular network. Under a similarity score threshold of 0.5, we compare Flink with six other spectra similarity algorithms: (i) DP (*33*) and (ii) entropy, which contain only fragment matching, (iii) hybrid similarity search (HSS) (*34*), (iv) bonanza (*26*), (v) GNPS, and (vi) entropy with neutral loss (entropy_NL), which consider both fragment matching and neutral loss matching. Flink algorithm displayed the lowest false-positive (0.7%) and a relatively high positive rate (90%) (Fig. 2C). Although the false-positive rates for DP (3.9%) and entropy (2.4%) were low, their positive rates were also low (65 and 67%, respectively). The other four algorithms that additionally considered neutral losses had high (>25%) false-positive rates. These results show that common neutral losses and fragments exclusion are essential for constructing our PFAS-only molecular network.

Building the PFAS_ID module

To automatic identify PFAS structure based on spectra connected by the PFAS_link module, we built a PFAS_ID module to transform the PFAS spectra into PFAS structure. The PFAS_ID module consists of two components: (i) one that calculates formula candidates from MS1 spectra and performs exact and transformation searches to retrieve candidate structures, and (i) another that uses a machine learning model to predict structure fingerprints from MS2 spectra. The fingerprints of retrieved candidates are then ranked based on their similarity with the predicted fingerprints.

We combined exact and transformation searches to retrieve candidate structures (Fig. 3A). Because of the lack of many PFAS in the structure database, the transformation search generated a combination of candidate parent SMILES and element transformations. The structure of the target SMILES was deduced based on this combination. To test the accuracy of the transformation search, we used 409 PFASs (*35*) that are absent from our in-house PFAS database and excluded isomers with the same formula for each test PFAS. We test the accuracy of transformation search in two aspects: formula accuracy, based on exact match of formulas between ranked candidates and target PFAS, and structure accuracy, based on the fingerprint similarity of candidate parent SMILES with target PFAS SMILES. Using predicted MS1 spectra (via the IsoSpecPy Python package), we retrieved formula and structure candidates for each of the 409 PFASs under a mass error tolerance of 5 ppm and an isotope threshold of 0.8. These candidates underwent ranking based on fingerprint similarity to the target PFAS and corresponding changes in elemental composition, as described in Materials and Methods. Among the 409 PFAS, the transformation search achieved a correct top one formula identification accuracy of 69%, under an average of 187 formula candidates. In contrast, this accuracy dropped to just 1.5% when randomly selecting formulas from the candidates without considering fingerprint similarity (Fig. 3B). In addition, in terms of structure similarity, the transformation search had an average top one fingerprint similarity of 0.76, while it was only 0.07 in random selection situation (fig. S3). Within the top 20 high-frequency transformations from the top 1 candidate for 409 PFASs, 12 were not in the Kyoto Encyclopedia of Genes and Genomes transformation database (*36*), including addition of F_2, CF_2, H_2S, and HF; loss of F_2, CF_2, and C_2F_4; substitution of F_2 by O or H_2; substitution of H by F or Cl; and substitution of F by H (Fig. 3C). These newly identified transformations suggest that transformation search can discover unknown PFAS through previously unidentified transformation pathways.

To predict structure fingerprints in the second component of the PFAS_ID module, we constructed a machine learning model using 194 PFASs spectra (table S1). We evaluated the structure diversity of our dataset by comparing it with PFAS datasets from a reference of previously identified PFAS (*35*) and a list of PFAS from the Organization for Economic Co-operation and Development (OECD). The comparison revealed that 94% of the structures from the reference dataset and 86% of the structures from the OECD dataset exhibited a maximum Tanimoto coefficient > 0.4 with our dataset (fig. S4). The average maximum Tanimoto coefficient was 0.75 for the reference dataset and 0.59 for the OECD dataset, indicating comprehensive coverage of structural diversity in our dataset (*37*). The PFAS structures were featurized using a combination of six types of substructure-related fingerprints, which amounts to 8034 bits of fingerprints (table S2) (*38*). Because most of the bits are unbalanced, we only retained 254 bits with positive rate from 10 to 90% after removing collinear bits based on the PFAS training dataset. Our analysis revealed that predicting these 254 fingerprint bits separately using 254 single-output models yielded higher accuracy compared to predicting all 254 bits collectively using a multi-output model (fig. S5). Featurizing the PFAS spectra by computing spectra similarity score with 194 compounds from the training dataset was more accurate and precise than direct featurization to mass vector (fig. S6). We compared how well support vector machine (SVM), logistic model, Bayesian model, decision tree model, random forest model, K-nearest neighbor model, and artificial neural network predicted the 254 fingerprint bits (Fig. 3E). All models except Bayesian had a median prediction accuracy above 85%, indicating that fingerprints can be accurately predicted from spectral information using machine learning models. Among the seven models, SVM had the highest median accuracy of 90.0% and median F_1 score of 81.8% (Fig. 3F). We also assessed the robustness of our model through 10-fold cross-validation, yielding a median accuracy of 90% with a SD of 1.5% (fig. S7). On the basis of these results, we trained 254 SVM models to predict structure fingerprints from spectra.

The final stage of the PFAS_ID module involves ranking candidate structures retrieved from MS1 spectra based on the similarity between their fingerprints and those predicted from MS2 spectra. To validate the performance of PFAS_ID, we used an external test dataset, which included 48 PFASs not in the training data, to compare our models with current identification platforms, CFM-ID 3.0, Sirius 5.5, and MetFrag 2.0. Their performances were evaluated by ranking candidates from PubChem database when the formula is known (Fig. 3G) or unknown (Fig. 3H). In the unknown formula case, MS1 spectra were given to the platforms but formulas were not. In our model, we used the parent ion mass and isotope distribution to calculate formula candidates and then retrieve structures from PubChem. CFM-ID used the same candidates as our model because it does not support PubChem database searching. When the formula is known (Fig. 3G), our model had the highest top one rate (66.7%), followed by MetFrag (56.3%), SIRIUS (50.0%), and CFM-ID (45.8%). When the formula is unknown (Fig. 3H), our model still had the highest top one rate (58.3%), followed by CFM-ID (43.8%), MetFrag (37.5%), and SIRIUS (12.5%) with a median number of candidates of 39. These testing results show that our method is most accurate at identifying PFAS.

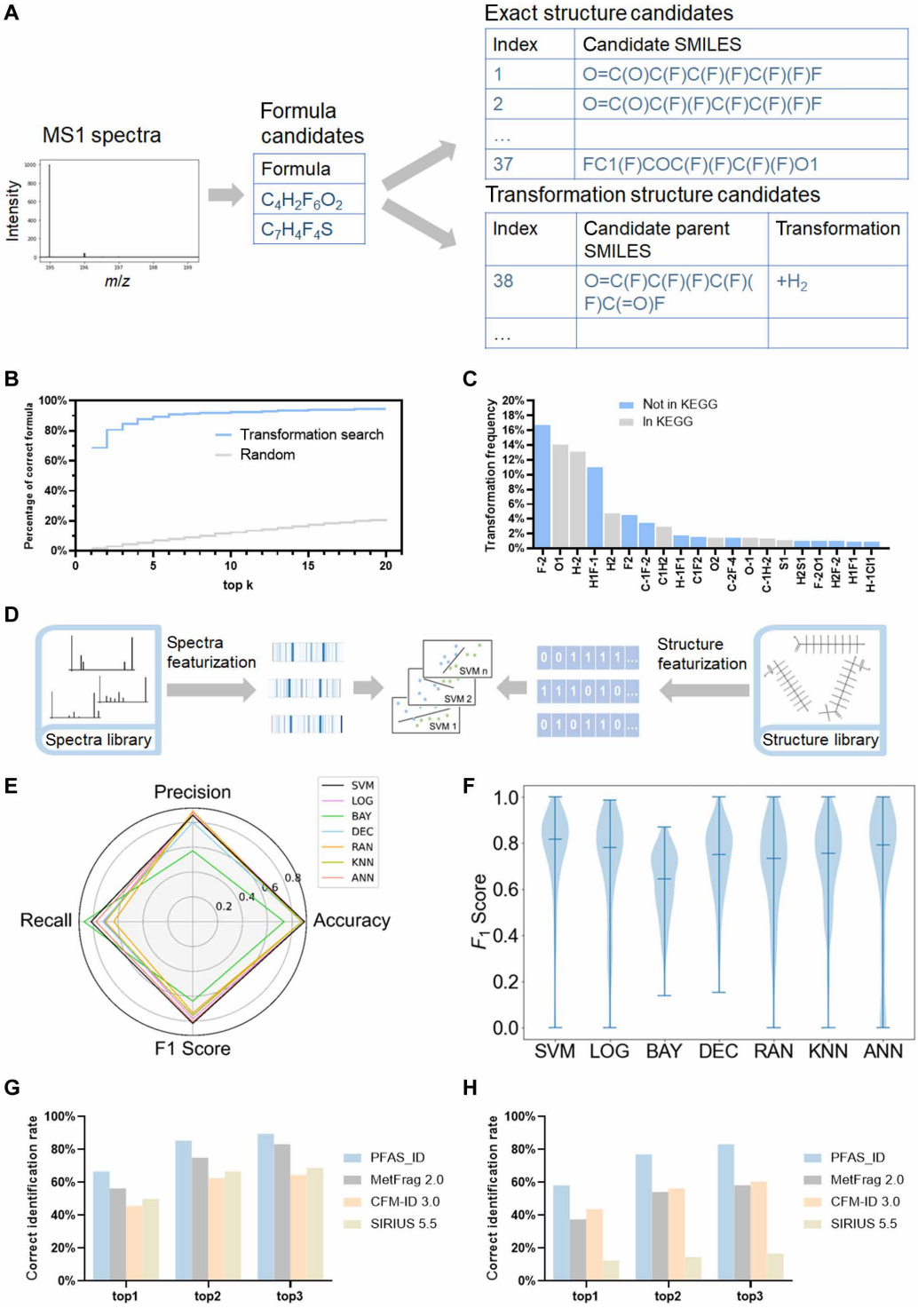


Fig. 3. Example and performance of PFAS_ID module. (**A**) Candidate search workflow example. (**B**) Correct formula rate of transformation search using 409 PFAS, random stands for formula randomly selected from formula candidates. (**C**) Top 20 transformation frequencies for transformation search on 409 PFAS. (**D**) Fingerprint prediction model training process. (**E**) Model accuracy, precision, and recall F_1 score comparison in predicting fingerprints on test dataset. SVM, support vector machine model; LOG, logistic model; BAY, for Bayesian model; DEC, decision tree model; RAN, random forest model; KNN, K-nearest neighbor model; ANN, artificial neural network. (**F**) Model F_1 score comparison in predicting fingerprints on test dataset. (**G**) Correct identification rate of PFAS_ID, MetFrag 2.0, CMF-ID 3.0, and SIRIUS 5.5, while formula is known in test set, correct identification rate represents the percentage of correct structures in top one, top two, and top three ranks for 48 test PFAS. (**H**) Correct identification rate of PFAS_ID, MetFrag 2.0, CMF-ID 3.0, and SIRIUS 5.5 while formula is unknown.

APP-ID identifies PFAS in fluorochemical wastewater

We used APP-ID to screen for unknown PFAS in wastewater samples obtained from a fluorochemical industrial park in Changshu, Jiangsu Province, China. The influent and effluent samples from the wastewater treatment plant were analyzed by liquid chromatography–HRMS (LC-HRMS). We used the sample data and PFAS MS/MS spectra database that contains 138 PFASs to construct a molecular network. All peaks were identified with our machine learning method and some were further confirmed by reference standards, spectral library, or through diagnostic evidence such as homologs. Following a previous study (*39*), confidence level 1 is assigned to matches by reference standards, level 2a for matches by spectral library, level 2b for matches identified by diagnostic evidence, and level 3 is for tentative structures. Adducts, insource fragments, coelution peaks, and redundancies were excluded.

APP-ID identified 431 seed PFASs in the influent after 14 rounds of iteration (Fig. 4A) and 442 in the effluent samples after 9 rounds of iteration (Fig. 4B). Initially, only 18 and 16 PFASs were identified in the influent and effluent samples, respectively, in round 0 using spectral library matching. The PFAS molecular network in the effluent was visualized in Fig. 4C, revealing that 80% of the PFASs were identified in rounds 0 to 4. Compared with matching spectral library, molecular network identified 24-fold more PFAS in the influent and 28-fold more in the effluent sample.

For example, from a network of 20 PFASs formed from a compound with peak ID C2917, we identified C28756 in round 3 from C4432 with a Flink similarity of 0.5 (Fig. 4D). The hydrogen, chlorine-substituted perfluoroalkyl ether carboxylic acid structure our model computed for C4432 matched the CO_2 + HF (mass error, 0.63 mDa) neutral losses in MS/MS and the structure retrieved from exact search of the structure database. In other automatic identification platforms, unknown PFAS not included in the structure databases escape detection. For example, exact search retrieved only two candidate structures (ranks 2 and 4) for C28756 but the neutral loss of CO_2 + HF (mass error, 0.68 mDa) cannot be explained in both structures (Fig. 4F). In PFAS_ID, the additional transformation search from our in-house PFAS database returned three candidates (ranks 1, 3, and 5). Rank 1 candidate with the highest score (166) is a reasonable structure because the carboxylic acid and hydrogen substitution matched the neutral loss of CO_2 + HF and the chlorine substitution matched the 3:1 isotope distribution of C28756. PFAS_ID also identified many CF_2 homologs of C28756 in the samples from C_7 to C_{16}. The retention time of this class steadily increased with precursor *m*/*z* (fig. S8), and both the MS and MS/MS spectra of this class had the same fragmentation pattern with diagnostic fragment $[M\text{-}CO_2\text{-}HF]^-$ (fig. S27). Besides, this PFAS class (ω-H,Cl-PFCA) had a similar fragmentation pattern with the synthesized standard α-H,Cl-PFCA ($C_6F_{13}CClHCOOH$), which displayed a diagnostic fragment $[M\text{-}CO_2\text{-}HF\text{-}C_4F_8]^-$ (fig. S10). Therefore, this PFAS class was assigned to confidence level 2b, and standards are still needed to confirm the exact structures. Notably, this class of PFAS has not been previously reported in environmental media.

In both influent and effluent samples, we identified 733 PFASs by APP-ID. A total of 130 PFAS structures from 31 classes, including perfluoroalkyl carboxylic acid, sulfonic acid, or alcohol, were identified in levels 1 to 3 by expert inspection (table S3). This included 15 PFASs from level 1, 18 from level 2a, 90 from level 2b, and 7 from level 3 (Fig. 5A). Class 1 are restricted legacy PFAS, classes 2 to 14 are emerging PFAS that have been identified previously, and classes 15 to 31 are unknown PFAS reported in this study.

Many of the unknown PFAS reported here have similar structures to known PFAS (Fig. 5A). For example, class 15 is derived from class 3 after the loss of F_2; class 16 is from class 8 after F is substituted by Cl; and class 17 is from class 8 after F is substituted by H. We verified the PFAS structures by comparing the MS/MS spectra for classes 1 to 14 with previous studies (*35*, *40*) and manually interpreting classes 15 to 31 by comparing them with known classes of PFAS. These unknown PFAS classes, which are absent from the PubChem database and cannot be identified using present platforms, were discovered through the PFAS database and transformation search on our APP-ID. The MS/MS spectra and fragment analysis of 31 PFAS classes were given in figs. S11 to S41.

Retrospective screening reveals global prevalence of PFAS

Although 733 PFAS features were identified by APP-ID in the fluorochemical wastewater, it remains uncertain whether these PFASs are present in other environmental samples. We retrospectively screened these PFASs using the Mass Spectrometry Search Tool (MASST) that searches for small molecule tandem MS data in all public metabolomics repositories (*41*). The screening returned 126 PFAS features consisting of 8 legacy, 37 emerging, and 81 unreported PFASs (Fig. 5B). More specifically, 44, 56, 6, and 20 PFASs were respectively detected in 1, 2 to 5, 6 to 10 and >10 datasets and 45, 27, 17, and 37 PFASs were respectively detected in 1 to 10, 11 to 30, 31 to 50 and >50 files (Fig. 5B). For comparison, we gathered reference spectra of PFHxS, PFOS, and PFOA standards, which are included in the Stockholm Convention, and they were found in 2 datasets (6 files), 7 datasets (205 files), and 14 datasets (121 files), respectively (table S8). These datasets could be categorized into seven sample types (Fig. 5C). Of the 126 PFAS features, most were found in environmental (86) and human (43) samples, illustrating the widespread existence of PFAS. Detailed information of screened PFASs were given in table S7, and examples of MASST matching results given in fig. S42.

Retrospective screening of the 126 PFASs against samples gathered from 20 countries further showed that emerging and unknown PFASs are found across the world, including Europe, China, India, Canada, Mexico, and Brazil (Fig. 5D). The United States recorded the highest count (113) of detected PFAS features because a large proportion of samples in the MassIVE dataset are from there. These results validate our methods but they suggest that the risks of unknown PFAS should be scrutinized and carefully reevaluated.

DISCUSSION

There are currently no platforms for identifying unknown PFAS in environmental samples. Platforms designed to identify metabolites and natural products cannot capture structurally different and increasingly diverse PFAS. To fill this need, we built an automatic PFAS identification platform by integrating molecular network analysis and a machine learning model. Molecular network based on spectra similarity algorithm has been used since 2012 to group natural products with similar structures (*42*). Although improved algorithms such as CSS (*43*), Spec2Vec (*44*), and MS2DeepScore (*45*) can transform spectra similarity to structure similarity, they often overlook the impact

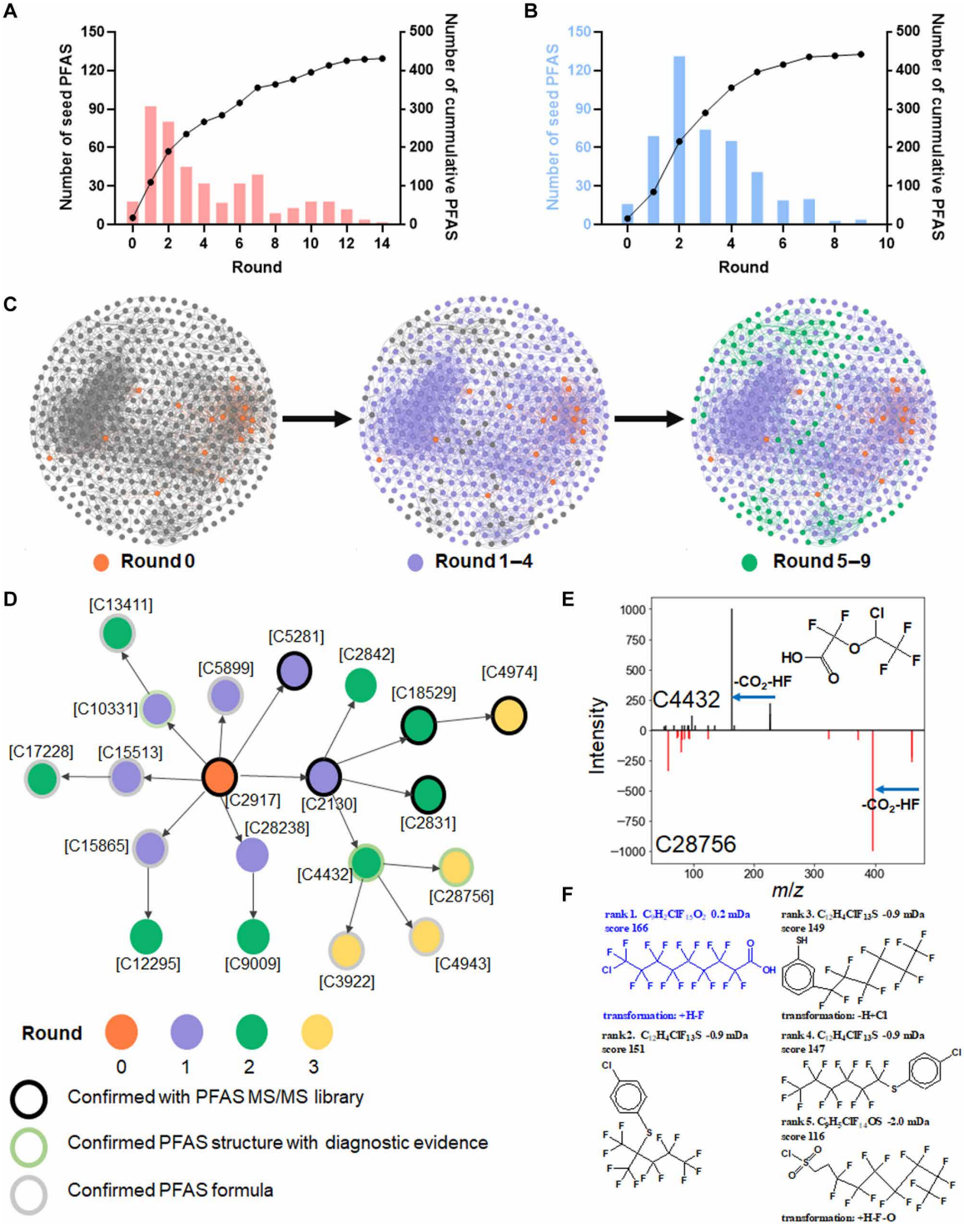


Fig. 4. Identification of PFAS in environmental samples. (**A**) Seed PFAS identified in each round from influent wastewater sample. (**B**) Seed PFAS identified in each round from effluent wastewater sample. (**C**) Network diagram of PFAS in effluent as an example to show the distribution of PFAS in round 0, rounds 1 to 4, and rounds 5 to 9. (**D**) An example to demonstrate the identification of 20 PFAS from C2917 in network of influent, where combinations of "C" and numbers represent compound peak IDs. (**E**) The MS/MS spectra of C4432 and C28756. (**F**) The ranked candidates of C28756 given by PFAS_ID model. The formula, parent ion mass error, and rank score of each candidate were provided.

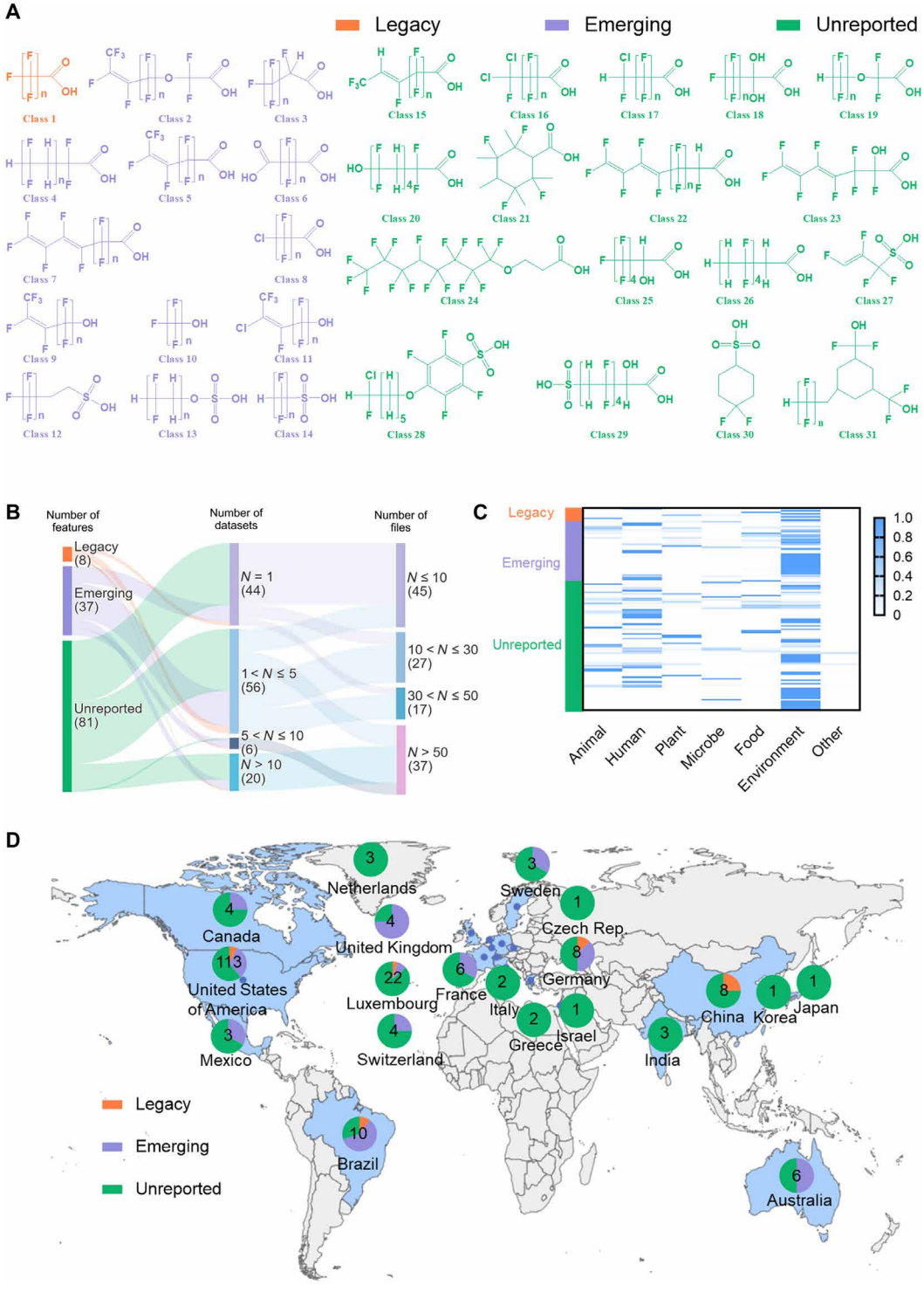


Fig. 5. Retrospective screening of PFAS using MASST. (**A**) Structures of 31 PFAS classes identified in fluorochemical wastewater. (**B**) Retrospective screening statistics for detected PFAS features in MassIVE datasets or files. (**C**) Distribution of PFAS features sample types using MASST retrospective screening. (**D**) Worldwide distribution of PFAS features using MASST retrospective screening.

of fragment size and elemental composition. Our study shows certain small fragments that are structurally irrelevant to PFAS can substantially affect spectra similarity. Therefore, we developed a Flink algorithm to exclude common fragments or neutral loss matches in our PFAS-only molecular network. This exclusion lowered the false-positive rate from 35 to 0.7% without affecting the positive rate. Fewer false positives mean that PFAS are identified faster and more accurately. Our Flink algorithm is applicable to other compound classes.

Molecular network technology enables the propagation from known compounds into the unknown ones (i.e., compounds not in the structure database). However, the process of identifying unknown compounds remains challenging and heavily dependent on expert experience. To discover unknown compounds, our platform uses transformation search to obtain parent structures and transformations for the unknowns. Unlike traditional transformation prediction tools such as EAWAG-PPS (*46*) or BioTransformer (*47*) that rely on known transformation pathways, our transformation search provides unknown transformations based on mass shift. For instance, H,Cl-PFCA could be identified by a mass shift of adding H and losing F from Cl-PFCA (Fig. 4F), while this transformation cannot be predicted by EAWAG-PPS or BioTransformer (fig. S44). This is particularly suited for discovering unknown PFAS, whose transformation in the environment or during industrial processes is largely unknown.

When compared with current platforms such as CFM-ID, MetFrag, and SIRIUS that identify known-unknowns (i.e., compounds that are in the structure database but not in the spectra database), our PFAS_ID module performed best whether the formula is known or unknown and even when the structure is absent from the database. Both MetFrag and CFM-ID tend to return many false-positive formulas because they consider only parent ion mass when retrieving formula candidates. SIRIUS, which retrieves candidates by fragment tree and Bayesian statistics, frequently excludes many true positive formulas, possibly due to the limited representation of PFAS in the training set (*48*). Except for CFM-ID that used structure candidates in our model, the identification rate for MetFrag and SIRIUS declined more sharply from PFAS_ID when the formula is unknown than when known.

When applied on fluorochemical wastewater, APP-ID identified 733 PFAS features, 130 of which were verified with high confidence. Spectra matching identified only 20 PFASs. Our results demonstrate that molecular network can effectively identify PFAS that are not in the spectra library. Using APP-ID, we discovered 17 unknown PFAS classes, 10 of which are single compounds that cannot be identified by traditional homolog screening. MASST is a powerful tool for screening compounds in diverse samples worldwide, allowing us to link compounds with sample location, type, time, and even certain diseases (*49, 50*). Therefore, the validity of sample information and the size of the dataset are vital for conducting more sophisticated analysis. Retrospective screening using MassIVE repository, containing environmental and human samples, returned 126 features comprising of eight legacy, 37 emerging, and 81 unreported PFAS from 20 countries. Our APP-ID, which can reveal more unknown classes of PFAS in human and environmental exposome than current platforms, is expected to support a more precise chemical management and risk assessment of forever chemicals.

MATERIALS AND METHODS

Chemicals

Forty-one authentic standards and 13 internal standards were used in this study, as summarized in table S4. The solvents and reagents used in this study are of high-performance LC grade, including methanol from Merck (Germany), water from Thermo Fisher Scientific (USA), ammonium acetate from CNW technologies GmbH (Duesseldorf, Germany), and trichloroacetyl chloride from J&K (China). Standards were used to test mass accuracy, isotope accuracy, the recovery of our sample pretreatment method, and the synthesis and confirmation of PFAS.

Sample preparation

Wastewater samples were collected in 2011 from one of the largest fluorochemical industrial park in China located in Changshu, Jiangsu Province. Influent and effluent samples were collected by peristaltic pump for 4 hours from the wastewater treatment plant of the fluorochemical industrial park. The wastewater samples were stored in 1 liter of polypropylene bottles under 4°C. We have filtered 300-ml influent and effluent samples, and internal standards were added after filtration, then the samples were concentrated to 1 ml in methanol by solid phase extraction using three kinds of cartridges; details are reported in our previous method (*13*). The samples were stored in −20°C before instrument analysis.

LC-HRMS analysis

The wastewater samples were analyzed using an LC system (Thermo UltiMate 3000, Thermo Fisher Scientific, USA) coupled to a quadrupole orbitrap MS (Q Exactive Focus, Thermo Fisher Scientific, USA). A C18 column (ACQUITY UPLC BEH C18, 2.1 × 150 mm, 1.7 μm, Waters, USA) was used for LC separation under 2 mM ammonium acetate water (phase A) and methanol (phase B) with a column temperature 40°C and an injection volume of 10 μL. The flow rate is 0.3 ml/min and the gradient was set as follows: 0 to 1 min 10% B, 1 to 36 min 10% to 100% B, 36 to 50 min 100% B, 50 to 50.1 min, 100% B to 10% B, and 50.1 to 55 min 10% B.

HRMS was operated with full scan (under mass resolution of 70,000) followed by data-dependent MS/MS scan (under mass resolution of 17,500) in electrospray ionization negative mode. Sheath gas, aux gas, and sweep gas were set on 48, 11, and 2, separately. Capillary temperature, aux gas–heated temperature was set on 320° and 413°C. Spray voltage was set on −2.5 kV. The S-lens radio frequency level was set on 50. Other settings were as follows: full scan range of m/z, 80 to 1000; full-scan automatic gain control (AGC) target, 1×10^6; MS/MS scan range, 50 to 1000; MS/MS scan AGC target, 5×10^4. Precursor ions were selected by quadrupole with an isolation window of 1.0 Da and fragmentation normalized collision energy of -35 ± 15 eV.

Peak picking and blank subtraction

We used MS-DIAL 4.18 software to pick peaks from both wastewater sample and blank control using its default parameters. Peaks with were aligned under a mass error of 0.01 Da and a retention time tolerance of 0.5 min using a Python script. Subsequently, peaks with CF_2-normalized mass defects (>0.85 or <0.15) were retained (*51*). Peaks with an area less than three times that of the blank control were removed, and peaks from internal standards were also excluded.

Annotation of adduct and coelution peaks

The adduct ion of $[2M\text{-}H]^-$, $[M\text{-}CO_2\text{-}H]^-$, and $[M\text{-}H]^-$ were annotated by a Python script. The mass error was set on 0.002 Da and retention time error was set on 0.5 min to annotate the adducts. Adduct ions of $[2M\text{-}H]^-$ and $[M\text{-}CO_2\text{-}H]^-$ were not used as seed PFAS in molecular network. Besides, coelution peaks were also annotated in a retention time error of 0.5 min and a DP spectra similarity threshold of 0.7. These coelution peaks were annotated to facilitate the removal of potential in-source fragments or coelutions from the results.

PFAS MS2 spectra library

A total of 242 MS2 spectra were obtained for 189 PFASs from various sources, including the AB SCIEX PFAS spectra database (60 spectra), PFAS spectra in MassBank (15 spectra), Norman PFAS spectra database (*35*) (122 spectra), in-house reference standard (20 spectra), and 26 PFAS spectra from other reference. In the training of the SVM model, 194 spectra from 141 PFAS structures were used. For the external test dataset, 48 spectra from 48 PFAS structures were used, ensuring no overlap with the training set (table S1). Sixty PFAS spectra from AB SCIEX were used in evaluating the molecular network performance, and 700 non-PFAS (fluorine free compounds) spectra were obtained in MassBank. In the application phase, we used a PFAS spectra database including 138 PFASs (58 spectra from AB and 80 from Norman).

PFAS molecular network

In the pretreatment of MS2 spectra, we filtered the spectra by removing all MS/MS fragment ions within ±12 Da of the precursor *m*/*z* and fragments with less than 5% relative abundance. For MS2 spectra matching, the fragment error was set at 0.01 Da. These settings were applied uniformly across all algorithms: Flink, DP, entropy, HSS, Bonanza, GNPS, and entropy_NL. In particular, we evaluated three scenarios for the Flink algorithm (Fig. 2A) and identified scenario 3 (excluding common neutral losses and fragments) as the optimal setting. Subsequently, Flink (scenario 3) was used in the construction of the PFAS molecular network. A similarity score above 0.5 and at least 1 matched peak are required to link two spectra in Flink. The Flink similarity score was calculated as follows

$$\text{Flink score} = \sqrt{\frac{N_{AB} * N_{BA}}{N_A * N_B}}$$

where N_{AB} is the number of match fragments/neutral losses from spectrum A to spectrum B, N_{BA} is the number of match fragments/neutral losses from spectrum B to spectrum A, N_A is the number of fragments in spectrum A, and N_B is the number of fragments in spectrum B.

Training of machine learning model

The machine learning models were trained using PFAS MS2 spectra database. The structure of PFAS were featurized into 254 fingerprints (table S2) as output; the spectra were featurized into a 194-dimension vector, while value of each dimension was calculated based on the Flink spectra similarity score with a training set PFAS. Spectra vectors were used to predict the 254 fingerprints separately and thus 254 models were trained. During the scoring phase, the total score for each candidate structure is calculated by multiplying its 254 dimensional molecular fingerprint scores with its corresponding F_1 scores and then summing them up.

Formula calculation and structure candidate retrieving

The formulas of potential PFAS peaks were calculated in Python script under the following setting: C: 2−50, H: 0−100, F: 0−50, O: 0−20, N: 0−5, P: 0−2; S: 0−2, Cl: 0−4; RDB was set in 0 to 20; relative mass error was set on 5 ppm; theoretical isotope distribution was calculated using IsoSpecPy Python package (*32*) and were compared with experimental MS1 isotopes given by MS-DIAL; and isotope threshold was set on 0.8, while isotope score was calculated as follows.

isotope score = $[1 - \text{abs}(M0_{exp} - M0_{mea})] * [(1 - \text{abs}(M1_{exp} - M1_{mea})] * [1 - \text{abs}(M2_{exp} - M2_{mea})]$. abs is for calculating absolute value; M0, M1, and M2 are for the relative intensities of [M] and [M + 1] and [M + 2]; exp. is for expected value; and mea is for measured value in sample.

Structure candidates were retrieved in both exact search and a transformation search. In exact search, we have built an offline PFAS structure database using PubChem including 6.95 million structure (excluding structures with fewer than three fluorine atoms, metal compounds, and stereoisomers) along with an in-house common PFAS dataset including 6484 PFASs from Norman (*35*), EPA dashboard (*52*), and OECD database (*53*). In transformation search, we used the 6484 in-house common PFASs as parent structure, and the transformation is calculated based on calculated formula and 6484 parent structure formula. The maximum number of transformation element was set at 10. The score of the candidates calculated in transformation search is calculated based on the parent structure score.

$$\text{Score}_{trans} = \text{Score}_{parent} * \left[1 - 0.02 * (K^2 + N)\right]$$

Score_{trans} is for the final score of this transformation search candidate; Score_{parent} is for the score of parent structure based on its fingerprint similarity with the predicted fingerprint; K is for the kind of element change in the transformation; and N represents the number of element changes in the transformation, with adjustments made for fluorine and hydrogen counts (multiplied by 0.5). Specifically, CF_2, CH_2, and CFH were counted as 1 in the N calculation. The parameters have been optimized using 6484 in-house PFAS data.

Retrospective screening using MASST

In MASST, a single spectrum search was completed using the online workflow (https://ccms-ucsd.github.io/GNPSDocumentation/) on the GNPS website (http://gnps.ucsd.edu). The data were filtered by removing all MS/MS fragment ions within ±17 Da of the precursor *m*/*z*. MS/MS spectra were window filtered by choosing only the top six fragment ions in the ±50 Da window throughout the spectrum. The precursor ion mass tolerance was set to 0.01 Da and a MS/MS fragment ion tolerance of 0.01 Da. The library spectra were filtered in the same manner as the input data. All matches kept between input spectra and library spectra were required to have a score above 0.6 and at least one matched peak (*54*).

Supplementary Materials

This PDF file includes:

Figs. S1 to S44
Legends for tables S1, S3 to S8
Table S2

Other Supplementary Material for this manuscript includes the following:

Tables S1, S3 to S8

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